

AMARNATH S. PATEL

Orlando, FL | amarnath.patel@ucf.edu | apatel.co | github.com/jeebuscrossaint

Physics and mathematics undergraduate — numerical methods, statistical inference, and performance-critical C++/CUDA.

EDUCATION

University of Central Florida B.S. Physics (Optics & Lasers) and Mathematics, Computer Science minor	Undergraduate Student 4.00 GPA August 2025 – Expected May 2029
• Mathematics: Linear Algebra (proof-based, in progress), Complex Analysis, Partial Differential Equations (in progress), Applied Linear Algebra, Ordinary Differential Equations, Mathematical Methods for Physics, Discrete Structures, Honors Calculus III • Computing: Data Structures & Algorithms, Computer Architecture, Computer Logic Design, Object-Oriented Programming, Deep Learning	
Florida Atlantic University 111 credit hours completed between ages 14 and 18, alongside high school	University coursework via FAU High School, ages 14–18 3.66 GPA August 2021 – May 2025

QUANTITATIVE RESEARCH EXPERIENCE

Independent Research — [CELERIS](#): high-performance numerical solver. C++23, CUDA, Python 2026

- Wrote a numerical solver for Maxwell's equations in periodic structures from scratch: Fourier-basis eigenvalue problems per layer, dense complex linear algebra, and a numerically stable scattering-matrix recursion chosen over the transfer-matrix formulation precisely because the latter loses conditioning as layers accumulate.
- Held the whole stack to measurable error bounds rather than eyeballed agreement — $1e-6$ against an independent method, $1e-15$ on multilayer recursion, $5e-12$ on a degenerate-case reduction, unitarity to 1.000000 — and cross-validated the finished solver against two third-party implementations to $\approx 1e-7$.
- Gradient-based (Adam) optimization over high-dimensional design parameters, plus Monte-Carlo tolerance analysis quantifying how performance degrades under manufacturing noise.
- Wrote the hot path as a CUDA kernel and benchmarked it against the project's own optimized 16-core CPU implementation: $4.8\text{--}5.8\times$ at 92k elements, agreeing to machine precision. Read the speedup's decline with problem size as evidence the kernel was memory-bound rather than compute-bound. *Sole-author manuscript in preparation.*

Undergraduate Researcher — [FAU Grant-Funded AI Safety Research Project](#) January 2024 – March 2025
Supervisor: Tucker Hindle, Florida Atlantic University

- Ran a signal-versus-noise study on a 255,000-passage corpus: benchmarked 10 candidate statistical discriminators (perplexity, transformer classifiers, latent semantic analysis, natural-language inference, burstiness) and ranked them by separation power. GPT-2 perplexity won at $3.35\times$ separation (17.5 versus 58.5); BERT next-sentence prediction turned out to carry no signal at all, which was worth establishing.
- Built four adversarial strategies against those discriminators — iterative rewriting, tree-search decoding, list-branching and reinforcement learning — over 3,125 generated sequences, and characterized a consistent tradeoff between evasion success and output quality. Grant-funded with HPC access.

Undergraduate Researcher — [UCF Astrophotonics Lab](#), CREOL August 2025 – Present
Supervisor: Dr. Stephen Eikenberry

- Estimation and signal processing on noisy experimental data: FFT sideband isolation and demodulation, Butterworth filtering, and a joint nonlinear optimization over three coupled parameters to recover complex amplitude and phase from raw interferograms — $\approx 98\%$ fidelity against simulated ground truth.
- Automated the four-instrument acquisition chain end to end, with in-loop optimization of a live objective and automatic rejection of bad frames, turning days of manual data collection into a single unattended run.

PROGRAMMING

- **Systems, in C, Rust and Go:** [hush](#), an interactive shell in C; [limebar](#), a Wayland status bar in C; [constrict](#), a build system in Go written to replace make; [hazmat](#), a password manager in C99.
- **Problem solving:** [Project Euler](#) and multi-year [Advent of Code](#) solutions in C++, plus [aocli](#), a command-line client for the latter. 1st place, Night Hacks hackathon.

SKILLS

Mathematics and methods: linear algebra and eigenvalue problems, complex analysis, differential equations, Fourier analysis, numerical optimization, Monte-Carlo methods, statistical benchmarking and discriminator evaluation, error analysis

Programming: C++ (C++23), Python (NumPy, SciPy), C, CUDA, Rust, Go; Linux, Git, CMake, GPU profiling

HONORS

Florida Bright Futures — Florida Academic Scholars (highest tier; 100% tuition)	2025
Lockheed Martin Award, “Highest Level of Engineering Excellence,” AEV Competition	2024
1st Place, Night Hacks Hackathon — UniUtils (course schedule generator and classroom finder)	2023